

Sterling Rooftop observatory manual

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General Rules.....	1
Opening up.....	1
Starting the telescope.....	2
Taking data.....	3
Slewing the telescope.....	4
Focusing.....	5
Shutting down.....	6
In case of emergency.....	7

General Rules

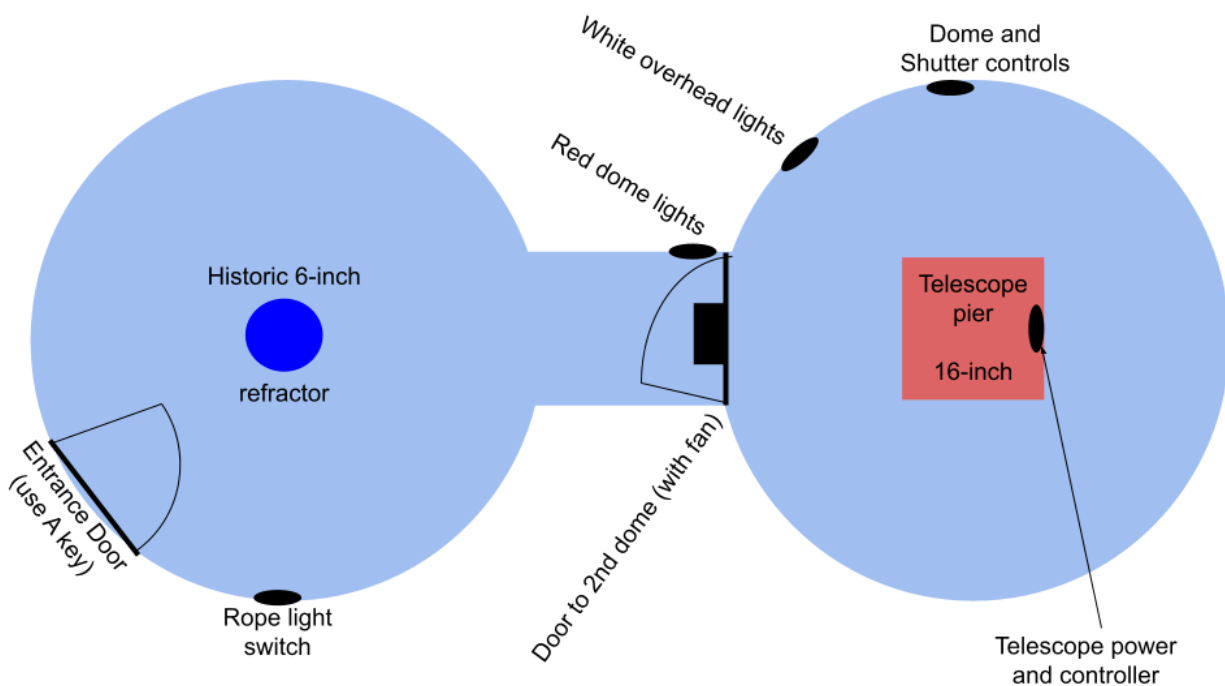
- SAFETY FIRST: Never observe alone, always have at least one other person in the dome with you at all times.
- Read the instructions first
- Please do not try to fix problems without an obvious fix yourself!
- If observing is interrupted for technical issues of any sort, please safely shut down the telescope and inform your instructor/TA. We will address the problem as soon as feasible.
- Make sure the weather conditions are favorable before opening up and at regular intervals throughout the night (see below)

Weather requirements

- No threatening clouds (thin clouds are ok, but if there is a chance for precipitation **DO NOT** open the dome)
- Outside temperature above 20 F
- Relative humidity below 90%, and no condensation on any surface (feel the outside of the dome or any of the exposed surfaces on the outside and inside. If any of them feel damp or wet this is a sign of condensation – do NOT open the dome, or close the dome if open)

Where is....?

Overview



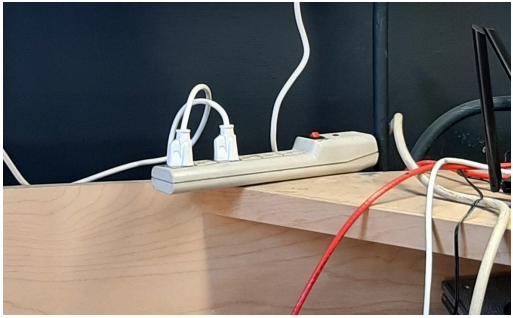
Light switches

Rope light switch

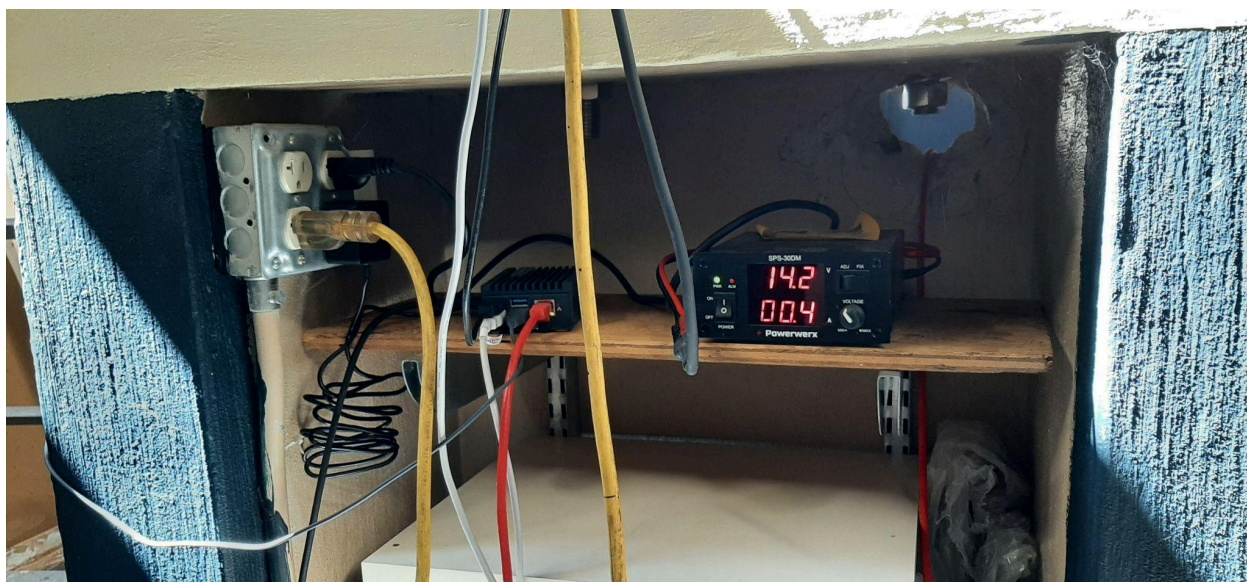
Located to the right of the main entrance.

These should be turned on during dark conditions to avoid trip hazards



Red dome illumination Just outside the door to the 2nd dome, in the little hallway connecting both domes	
White lights in dome Turn on via the switch on the power strip located on the upper shelf on the left hand side after entering the 2nd dome. White overhead lights can be turned on for setup and during maintenance, but must be turned off during night time observations.	

Telescope power and controller



The power to the telescope and all instrumentation is located on the west side of the pier, opposite side to the entrance.

These include:

- A yellow extension cord providing power to the instruments. Make sure this cable is plugged in when the telescope is in use, and also ensure it is **unplugged** when the telescope is **NOT** in use.
- Powerwerx power supply to the telescope: Turn on via the switch to the left of the display. Voltage should read 14.2V. Do NOT change the voltage, as this may result in permanent damage to the (very expensive) mount.
Turn power OFF when the telescope is no longer in use.
When the telescope is powered on the backlights on the keypad should also be illuminated.
- The telescope and all its instruments are controlled via a Raspberry PI controller shown above between the power outlet and the powerwerx power supply. This controlled must be running at all times. Before observing, ensure that the Raspberry PI is plugged in and running (check for the LEDs being lit up and/or blinking), that the red network cable is connected (check for a green LED), and the black and white USB cables are connected (black is for instruments, white goes to the mount).

Dome control

Dome control includes two different motors: (1) for opening the dome shutter, and (2) to rotate the dome. Both have separate motors but share a single outlet, so depending on which motor you want to operate the cable setup will look slightly different:

Open/Close Shutter



To open/close the shutter, plug in the plug labeled “Shutter” into the outlet. You also need to connect the square connector hanging down from the dome. This connector can only be connected in one orientation.
Note: If shutting down, make sure to put the dome in the correct position by lining up the two red arrows.

Rotate



To rotate the dome, disconnect the square connector to the shutter motor hanging down from the dome, and plug in the plug labeled “Rotator”

To open/close the shutter use the left controller to operate the motor. The motor stops automatically once the shutter is fully open/closed. Make sure to turn the motor control lever to idle once the motor has stopped.

To rotate the dome use the right lever switch. Make sure to allow the dome to come to a full stop before changing directions.

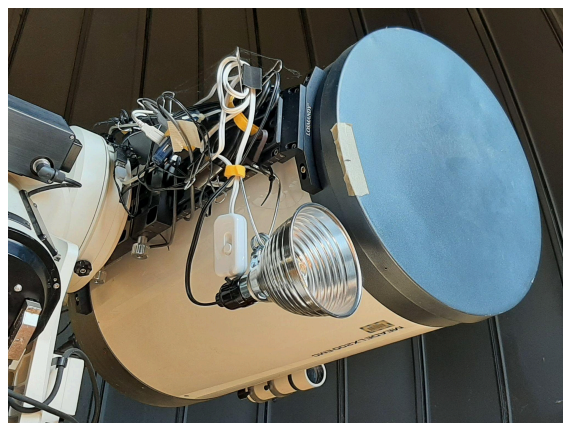
Instrumentation



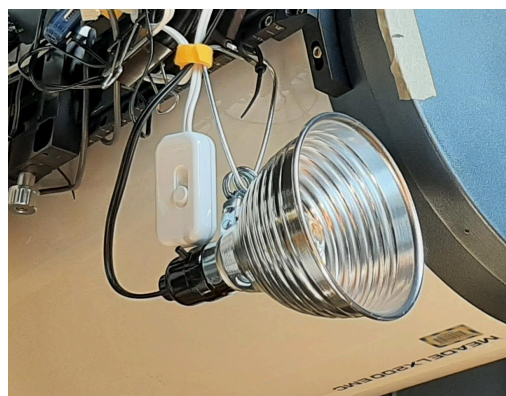
Telescope in its Park Position – whenever the telescope is not in use it should be parked in this position; optical tube horizontal, facing south towards the flatfield spot, with the counterweights also being horizontal pointing west, i.e. in the direction of the door.

Telescope

including installed mirror cover

**Flatfield lamp (with dimmer)**

To turn on/off the flatfield lamp push the small black button at the base of the lamp. The white dimmer can be used to change the lamp brightness (only do this is really necessary, it's a lot easier and more reproducible to instead vary the integration times)

**Telescope keypad**

CCD Camera

The blue device at the very end is the CCD Camera; it should be connected with two cables, a black power cable and a off-white USB cable.

When powered on you should be able to see/hear the four cooling fans spinning.

**Filter wheel**

Directly in front of the CCD is the QHYCCD filter wheel. The filter wheel is connected via two cable, a thinner power cable, and a slightly thicker grey USB cable for communication.

To check if it's operational verify that the LED installed on the telescope-facing side above the grey USB cable is lit up.



Opening up

1. Open the door to the first dome, and prop it open with the wooden chair located inside the dome. Keep this door open for both airflow and so you don't lock yourself out
2. Turn on the red floor lights (see above). The switch is located about 2 feet to the right (as you walk on) of the door. Leave those on for safety reasons, they won't bother the observations
3. In the 2nd dome, turn on the red LED lights – switch is located on the left wall just before the middle door
4. Optional, but encouraged for startup: Turn on the main lights in the 2nd dome. This is done via the switch on the extension strip located on the top shelf to the left of the door as you walk in.
5. Open up the dome shutter and prepare dome:
 - a. on the north wall of the dome there's a contraption with two cables and two plugs. Locate the one labeled "Shutter" and plug it into the single wall outlet nearby.
 - b. If not already done, plug in the 8-pin connector connecting the wall plug with the cable to the shutter motor that is dangling from above
 - c. Flip the left switch labeled "Shutter" and "Open/Close" to open up the dome shutter. Once open, return switch to neutral position
 - d. Unplug the 8-pin connector cable
 - e. Unplug the "Shutter" plug from the wall outlet and instead plug in the other cable labeled "Dome rotation"
 - f. You are now ready to rotate the dome using the 2nd flip switch to the right of the dome shutter switch.

Starting the telescope

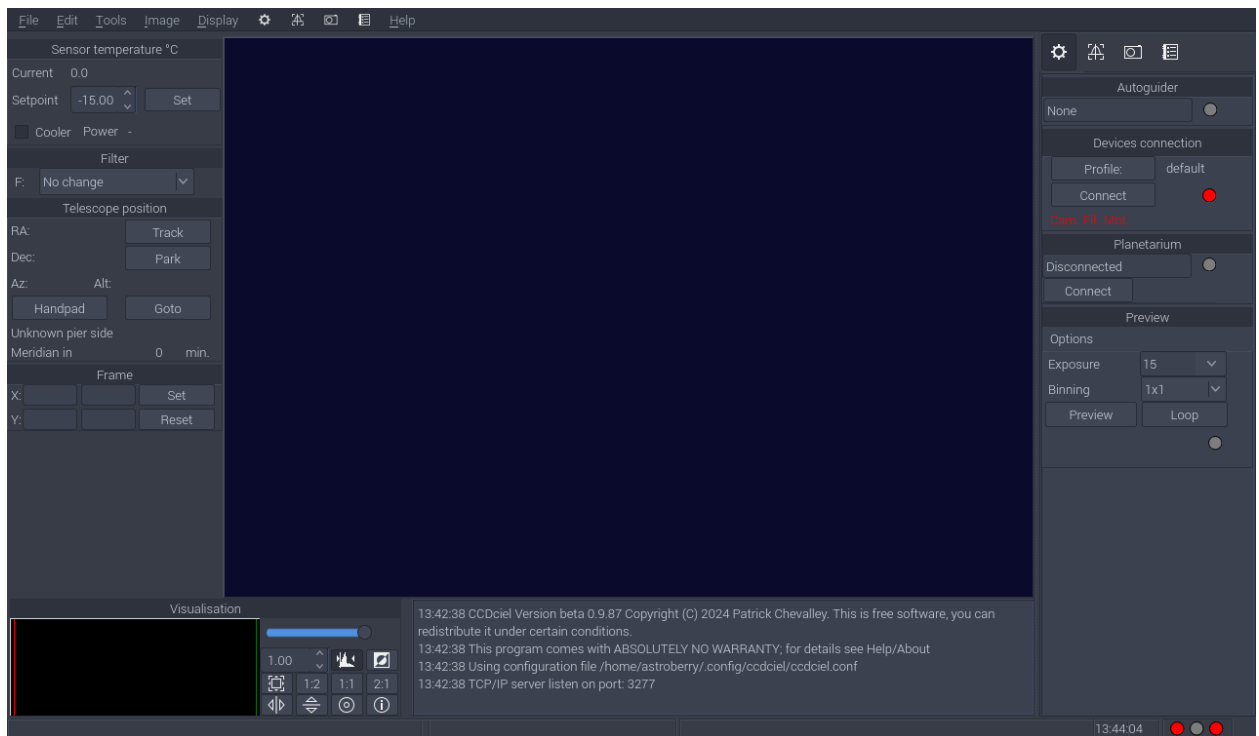
1. Remove the mirror cover from the mount and put it in a safe spot. Make sure to not move the telescope by hand.
2. Power on the mount
 - a. Locate the power supply on the east side of the pier (i.e. opposite side to where you came in from)
 - b. Turn on the power switch on the left (please leave the other switch on the right alone; this allows to modify the voltage supplied to the mount)
 - c. Check the Keypad display – it should start up and display the standard menu
3. Turn on the instruments
 - a. There is a power cord coming down from the extension strip mounted on the telescope. Connect this to the yellow extension cable coming from below that has a light-up plug.
 - b. Once connected you should start to hear the cooling fans on the CCD camera
4. Locate the Raspberry PI controller located next to the telescope power supply box (underneath the pier) – look for a small box in a clear housing with a bunch of cable connections. It should be connected to 2 USB cables (one for the mount, one for the instruments), 1 ethernet cable, and a micro-USB cable that supplies power. If the controller was not powered up before wait for it to boot up, this should take less a minute
5. Start up the laptop you will be using to take the controller. You can connect this to either the usual UWNNet or eduroam, or to a network named “astrodome” – password is “observer” without the “” (Please keep this password private, I'm not sure what UW would think about this setup).
6. Connect to the Raspberry PI via a web-browser:
<http://astroberry.astro.wisc.edu/desktop>
This will ask you for a password to log in; password is **astroberry**
Note: If getting to this IP address doesn't work try again after connecting to the astrodome WiFi. It should be accessible from the department network, but may not be available via the general UWNNet/eduroam connection.
7. On the Astroberry, open a terminal, and run
`./start_indi.sh`
This will start the software that handles all the behind-the-scenes communication between mount, camera, filter wheel, etc. You can minimize this window, but make sure to keep it open.

Taking data

1. Open CCDciel – in the start menu (looks like a planetary system in the top left), go the “Education”, and then click on CCDciel

Note: For whatever reason CCDciel sometimes shuts itself down. In that case simply open it back up and continue with the rest of the instructions here

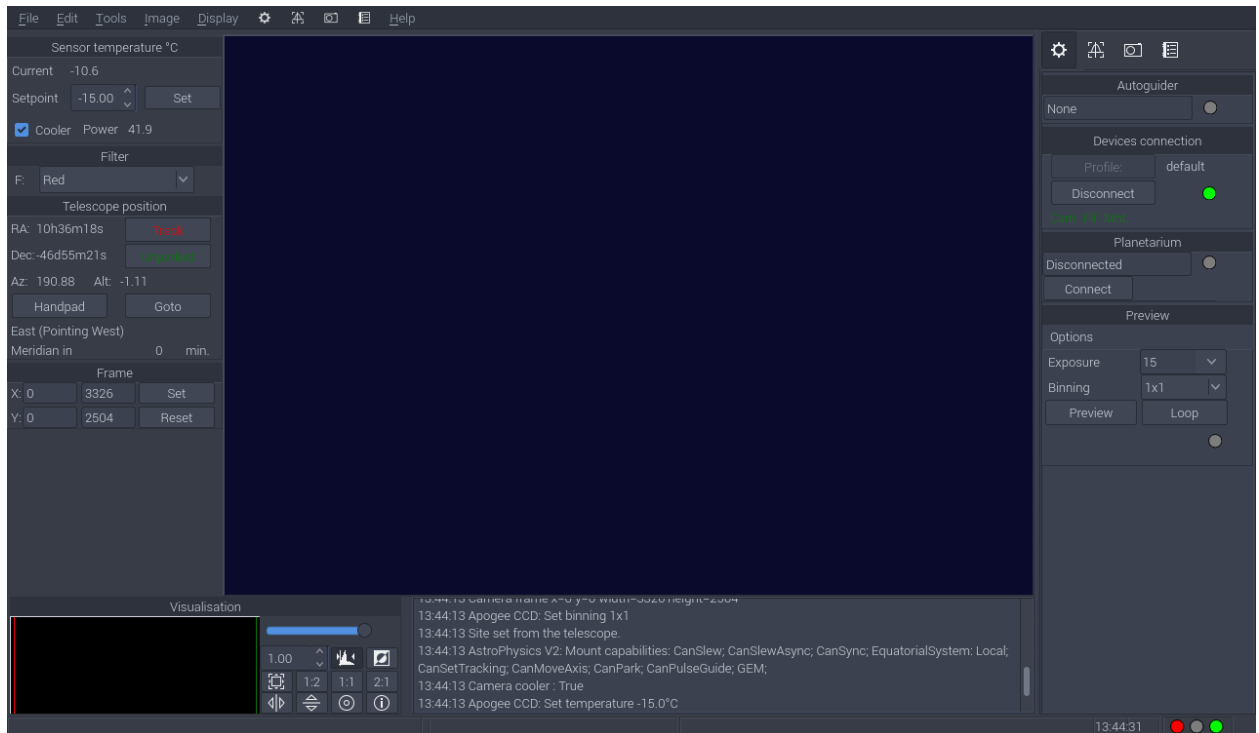
2. On the right you will find a button labeled “Connect” – click it to establish the connection between CCDciel and the rest of the instruments



- Wait for the three options below to turn green – it should look like this

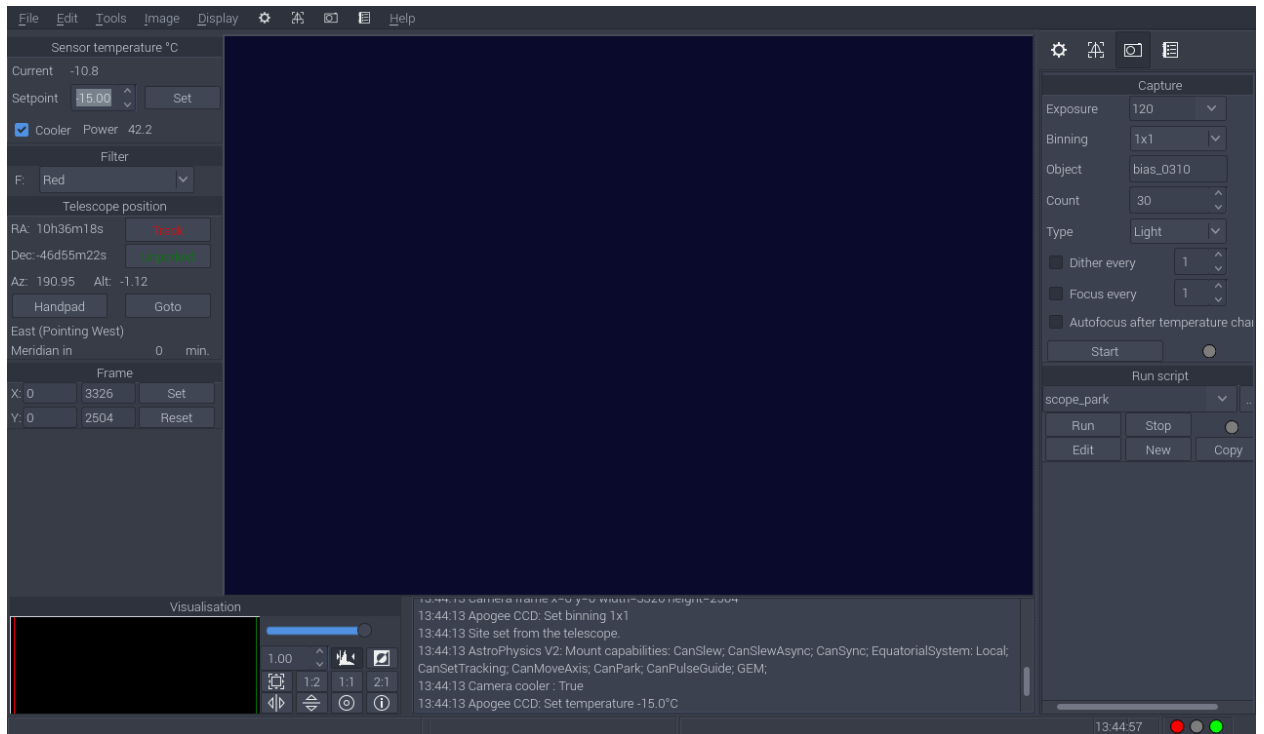
Mount Filt Camera

If one or more of them remain grey or red make sure start_indi.sh is still running in the terminal



- On the left of the window turn on camera cooling. Enter the set point (-10 or -20 should work well), and click the “Set” button. Also make sure the “Cooler” checkbox is activated.
NOTE: We currently have some condensation problems inside the camera, causing frost and ice to form on the detector when cooling is set below freezing. For the time being, either leave the cooling turned off or use a set point no lower than +10 deg.

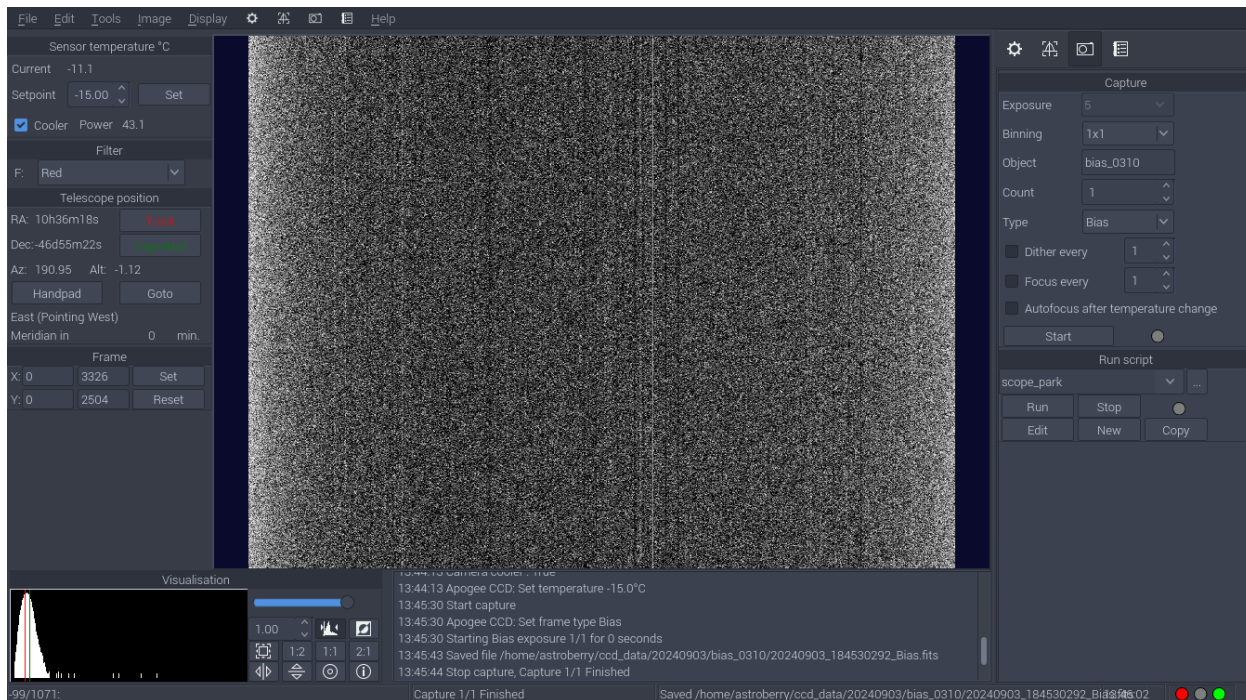
5. To take actual observations, select the “Camera” button in the right sidebar menu



Select the required settings for

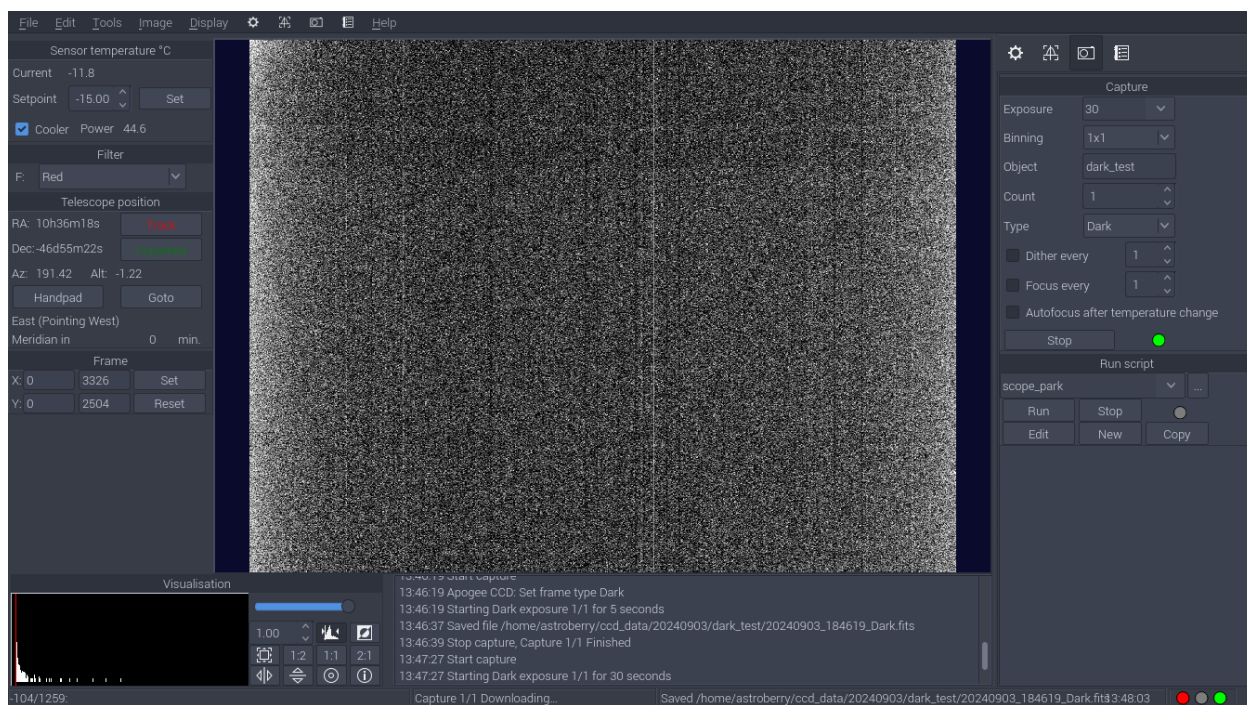
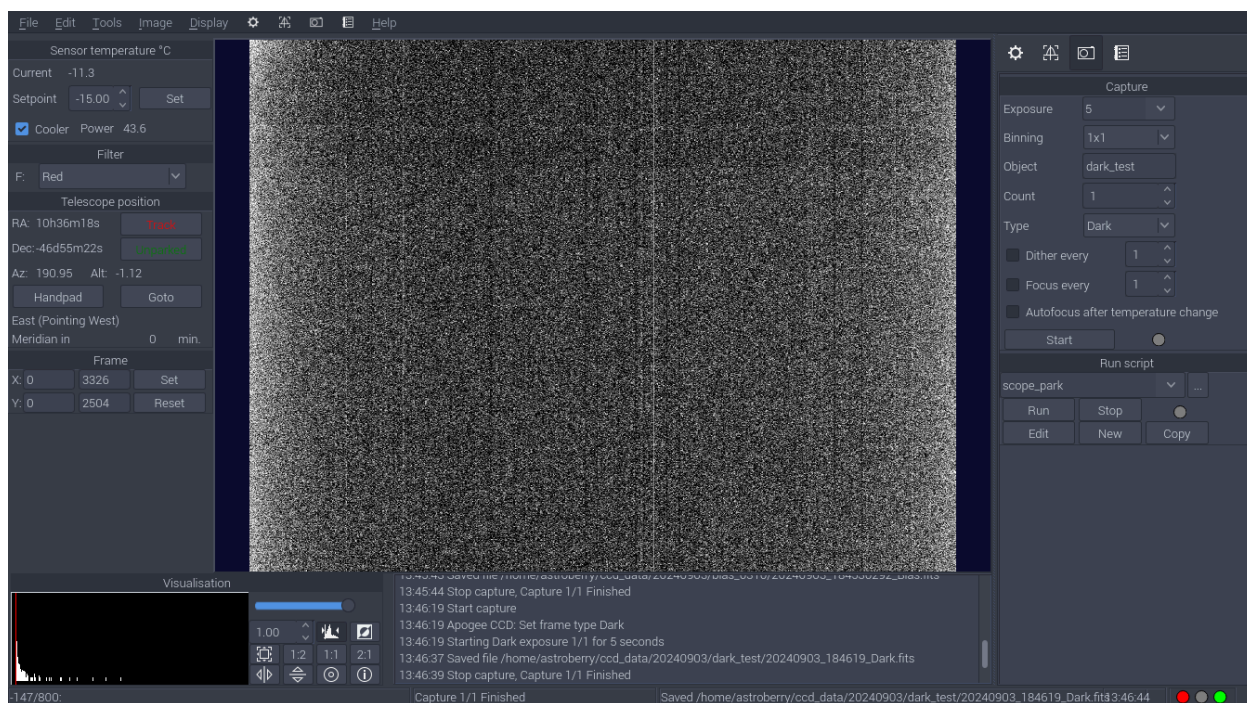
- “Exposure” (that’s the exposure time)
- “Binning” (leave set 1x1 for science frames, but can go to 2x2 or even 3x3 for check images; that way readout times are lower)
- “Object” gives some description to what you are observing and also will be saved as part of the filename
- Count – how many repeat observations
- Type – select from “Light” for science images, “Flat” for flatfields, etc.

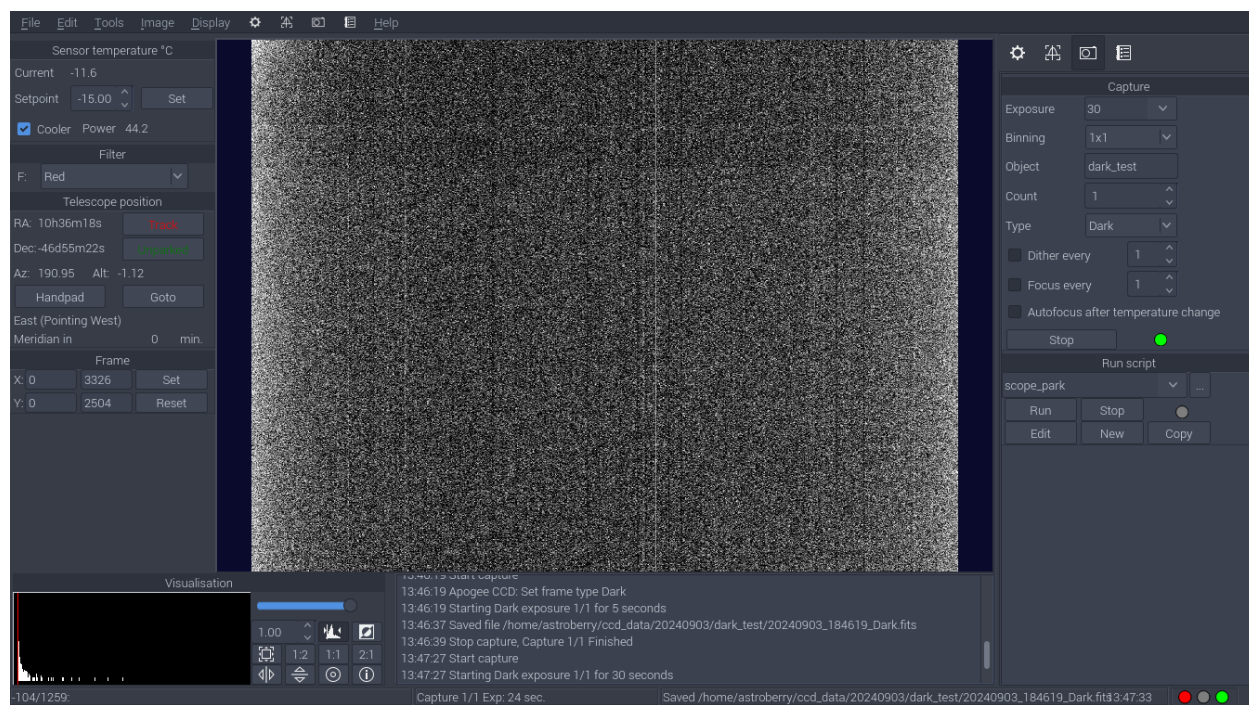
- At the start of every observing session make sure to take at least a few biases for reference. These will be required during data reduction, but also serve as a cross-check to make sure the instrument is operating normally:



If the bias image does not show any structure (except some gradients towards higher fluxes near the left and right edge) that means all is good.

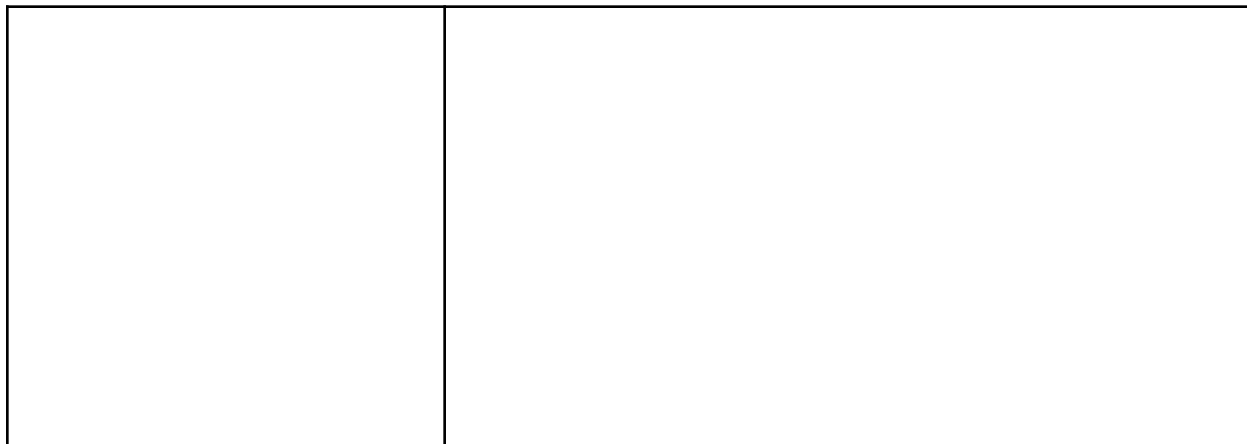
- For longer integrations you may also want to obtain dark observations.
- Currently tracking precision is a limiting factor for integration times. Depending on where on the sky your target is, try to limit individual integrations to 60 seconds to avoid star trailing. If you notice that your stars are still nice and round in a 60 second image feel free to go longer.





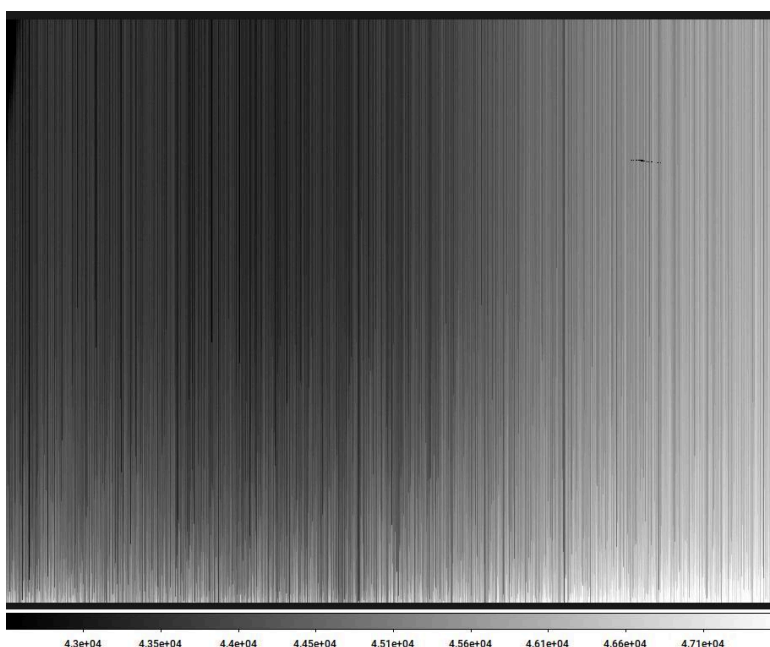
Help, my images look weird

Starfish illumination pattern in flatfields: exposure times too short



Streaky flatfields: exposure times too long

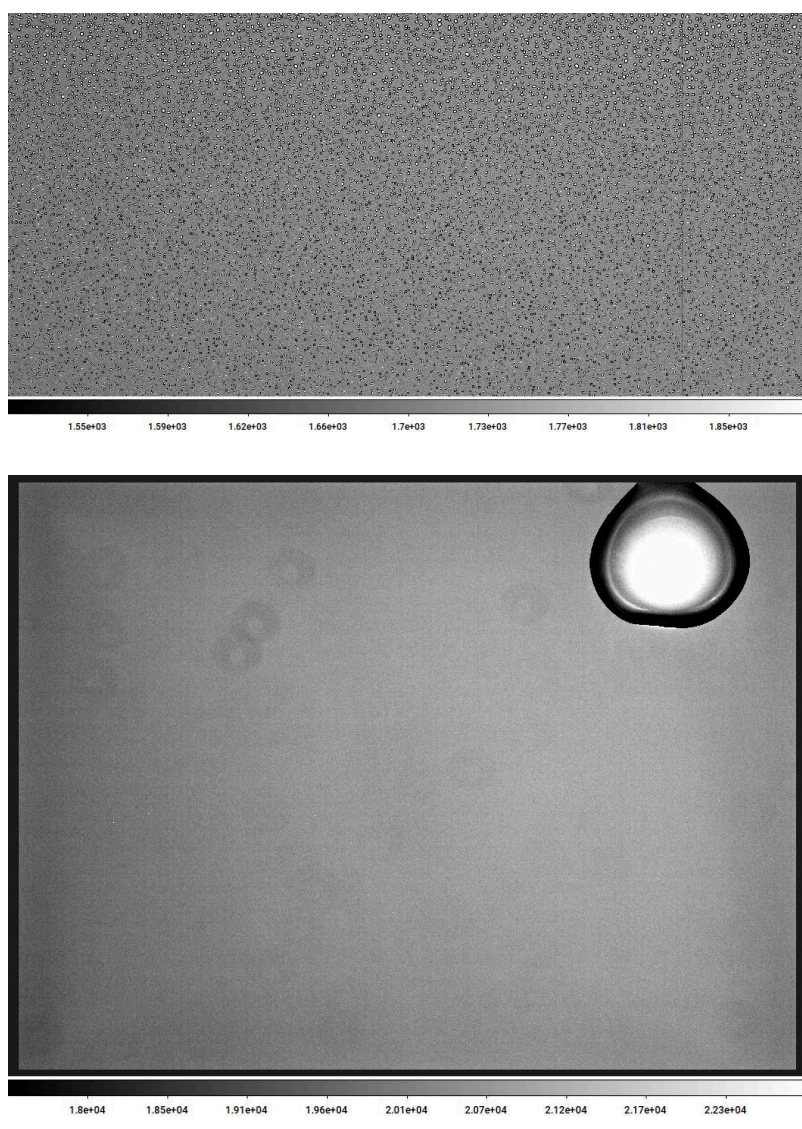
If your exposure times for your flatfields are too long you saturate the detector, ending up in data that looks “streaky”, and is no longer useful for flatfielding. This seems to happen around count levels of ~40,000 counts, so best try to stay at levels closer to or below 35,000 counts:



Spotty images

Spotty flatfield images – i.e. images with lots of bright (or dark) spots all over, or with fewer larger spots indicate the presence of frost and/or condensation on the detector.

To fix this, increase the CCD cooling set point (or turn off cooling altogether, and give it some time for the condensation to evaporate).



Slewing the telescope

Changing targets can be done in one of two ways, via the software (recommended) or the Keypad attached directly to the telescope

1. In the software, locate the “Goto” button in the left sidebar. This will open up a new window
2. Enter your Target name, and click “Search”. This should populate the rest of the fields with the resolved RA and DEC coordinates. If so you can skip the next step
3. If the automatic name resolution does not work you can enter your coordinates manually. Format for RA is HH:MM:SS, and +DDD:MM:SS for declination.
4. Once some value for both is populated it will automatically calculate an Azimuth and Altitude for your target. Once you are happy click on “Goto”
5. **IMPORTANT: Please keep an eye on the mount and telescope while the telescope is slewing. Watch out for any obstacles, cables, etc to avoid damage to any of the components.**
6. Rotate the dome so the telescope has a clear, unobstructed view of the night sky
7. By default, once the telescope has acquired its target it will go through some refinement steps, so be patient. These include
 - a. First a small delay to give the dome a chance to catch up and all motion besides the tracking to stop
 - b. A short 15-second exposure in the r-band
 - c. Once complete, the software will try to match star positions with known stars to determine where exactly the telescope is pointing.
 - d. If successful it will fine-tune the pointing of the telescope to bring the selected target near the center of the field of view
 - e. **HINT: As first target of the night select a field near the plane of the Milky Way that has a good amount of stars in it. That way the telescope pointing should be improved for all subsequent exposures.**
8. Once the fine slew is complete the telescope automatically starts tracking, so you can start taking data

Focusing

If you notice that stars in your data look like giant fuzzballs or donuts you likely have to tweak the focus of the telescope. For now this is a somewhat tedious, manual process. **Since it involves stepping on the ladder at night it strictly requires at least two observers in the dome!**

1. Locate the focus knob on the back of the telescope
2. Turn the knob $\frac{1}{8}$ of a turn in one direction. Let go of the telescope and give it a few seconds to settle down.
3. Take another exposure and see if the focus improved and your images became sharper. Hint: If you select binning of 2x2 your read-time will be quite a bit shorter. But do NOT mistake the smaller stars in the binned data with a better focus.
4. If so, repeat steps 2 & 3 until you are happy with the focus
5. If image quality become even worse, try turning the knob in the other direction.
6. Go SLOW: The focus knob directly moves the primary window, so small tweaks go a long way. Do NOT turn the knob too far before taking more data.

Shutting down

At the end of the night or when you are done with observations:

1. **Make sure to take your data home; DO NOT REMOVE YOUR DATA OFF THE PI.**

Options for doing so are:

- a. Upload it from the PI directly to Box/GDrive/elsewhere
 - b. SCP it from the PI to somewhere in the department network
 - c. Grab it from the PI using FTP – either from the command line or some FTP GUI software on your laptop. Connect to ftp://144.92.179.207. Username is “anonymous”, and any password will do in that case.
 - d. wget is another option that works well and should come standard with linux and Macs. command line you need is:
`wget -m -np -nH 144.92.179.207/20231123`
(replace the last part with your night)
 - e. Data is organized into directories labeled by the observing night, so e.g. 20231123 for November 23rd, 2023.
 - f. This is experimental, but if you bring a USB stick you should be able to copy the data onto your stick as well.
2. Park the telescope, either using CCDciel or the keypad (use Parking position 4). Wait until the telescope motion stops. Final location should be with counterweights being horizontal and pointing towards the door, and telescope pointing towards the horizon facing south (or towards the right if seen from the door), and CCD camera pointing towards the security camera
 3. In CCDciel, In the “Device Connection” Tab in the right sidebar, disconnect the software from controller by clicking “Disconnect”
 4. Unplug the instruments from the yellow extension cord. The fans in the CCD should stop
 5. Rotate the dome so the dome shutter is facing south and the little red arrow hanger lines up with the indicator on the dome.
 6. Disconnect the dome rotator plug from the wall outlet.
 7. Connect the 8pin connector with its receptacle hanging down from the dome shutter motor
 8. Plug in the “Shutter” plug into the wall output
 9. Close the shutter until it stops moving. Then return the switch to the neutral position.
 10. If the Raspberry PI is plugged into a wall outlet: leave it on.
If it's powered through the laptop: disconnect it from the laptop
 11. Please take the observing laptop with you and put it in the 3rd floor observing lab (alternatively you can put a post-it label with “Rooftop/REU” on it and leave it in the mail room)
 12. Make sure to turn off all the lights in the dome, and take all your belongings with you
 13. Turn off the red rope light via the switch in the first dome
 14. Put the chair that blocked the door open back on the inside and out of the way
 15. Close down the first dome

What data do I need to take?

At the very minimum take these data:

- 1) A set of a few bias frames (sometimes also referred to as zero frames). These are a readout of the detector with a zero exposure time and without opening the shutter. Why a few? During data reduction you'll combine individual bias frames into a master-bias to reduce noise.
- 2) Flat-fields for each filter you are using for your science data. Due to the particular design of the shutter in our CCD it's advisable to use domeflats rather than sky-flats. Without changing the lamp setting or filter, take a few flats with very short integration time (~0.1-0.2 seconds; note: these will be underexposed), as well as a set of flats with a longer integration time. The goal is to accumulate enough signal for good signal-to-noise ratios, but without saturating the detector. A good exposure level to aim for is 30-40K counts.
- 3) FOCUSING!!! This is not data you will use for the analysis, but it's essential to make sure the data you are taking is well focused. If not, photometry (i.e. measuring brightnesses) will be less accurate, your stars will look big and puffy (or even like donuts if the focus is really bad), and you won't be able to detect faint sources.

Your science images. Make sure to select a binning setting beforehand, and use the same binning setting for all science AND calibration data. 2x2 or 3x3 seems to work pretty well given typical observing conditions from atop a building in the middle of campus in downtown Madison. Observing times should be selected long enough to yield sky-noise limited observations (i.e. >15 seconds), but not too long to avoid saturating the detector or cause elongated stars due to tracking limitations (i.e. < 120-300 seconds or so based on location on sky)

In case of emergency

Please do not try to fix problems without an obvious fix yourself!

General rule is, if observing is interrupted for technical issues of any sort, please safely shut down the telescope and inform your instructor/TA. We will address the problem as soon as feasible.

For real emergencies (i.e. dome won't close and it's starting to rain), call Ralf Kotulla – 414-678-9365